

## Sec-6: Agile User Stories & Requirements Traceability

### 6.1 Introduction

This project was developed following **Agile methodologies**, specifically a tailored Scrum-Kanban hybrid suited for a solo developer context. Work was organized into **two-week sprints** with a prioritized product backlog maintained throughout the development lifecycle. Each feature was decomposed into **vertical slices** spanning the full stack — from Razor views and CSS styling down to database migrations and service-layer logic. The user stories below represent the complete functional scope of the e-commerce platform, organized into logical **Epics**. Each story includes a **priority rating** (High / Medium / Low) and **acceptance criteria** expressed as concrete, testable conditions. These stories collectively trace back to every controller action, service method, database entity, and view component in the system.

### 6.2 Epic 1: User Identity & Security

| ID     | As a...            | I want to...                                                             | So that...                                                                | Acceptance Criteria                                                                                                                                                                                                                                                                                                                                                                                                                                       | Priority |
|--------|--------------------|--------------------------------------------------------------------------|---------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| US-001 | New visitor        | Register a new account with my name, email, phone, address, and password | I can access personalized features like cart, wishlist, and order history | <ul style="list-style-type: none"><li>• Registration form validates all required fields (<b>FullName</b>, <b>Email</b>, <b>PhoneNumber</b>, <b>Password</b>)</li><li>• Password must meet Identity policy (6+ chars, digit, lowercase, uppercase)</li><li>• Duplicate email returns a validation error</li><li>• On success, user is signed in automatically and redirected to Home</li><li>• A welcome email is sent to the registered address</li></ul> | High     |
| US-002 | Registered user    | Log in using my email and password                                       | I can resume my shopping session                                          | <ul style="list-style-type: none"><li>• Login form validates email format and non-empty password</li><li>• Invalid credentials show a generic "Invalid login attempt" message</li><li>• Successful login redirects to the originally requested page or Home</li><li>• Cookie is set with 30-day sliding expiration</li><li>• Locked-out users see an access-denied message</li></ul>                                                                      | High     |
| US-003 | Authenticated user | Log out of my account                                                    | I can securely end my session on shared devices                           | <ul style="list-style-type: none"><li>• Logout button is available in the user dropdown menu</li><li>• POST to <b>/Account/Logout</b> clears the authentication cookie</li><li>• After logout, user is redirected to Home</li><li>• Protected pages (Cart, Checkout, Profile) are no longer accessible</li></ul>                                                                                                                                          | High     |

| ID     | As a...                   | I want to...                                                   | So that...                                             | Acceptance Criteria                                                                                                                                                                                                                                                                                                                                                                   | Priority      |
|--------|---------------------------|----------------------------------------------------------------|--------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| US-004 | <b>Registered user</b>    | Reset my password if I forget it                               | I can regain access to my account                      | <ul style="list-style-type: none"> <li>• "Forgot Password" link is visible on the Login page</li> <li>• User enters their email and receives a password-reset link</li> <li>• The link contains an Identity-generated token and expires after 1 hour</li> <li>• Token reuse is prevented (single-use)</li> <li>• New password must satisfy the same policy as registration</li> </ul> | <b>Medium</b> |
| US-005 | <b>Authenticated user</b> | View and edit my profile (name, phone, address, city, country) | I can keep my shipping information up to date          | <ul style="list-style-type: none"> <li>• Profile page pre-fills all fields from the database</li> <li>• Changes are persisted via <code>userManager&lt;ApplicationUser&gt;.UpdateAsync()</code></li> <li>• Email is read-only (cannot be changed)</li> <li>• Success/failure messages are displayed via TempData</li> </ul>                                                           | <b>Medium</b> |
| US-006 | <b>Authenticated user</b> | Change my password from within my account settings             | I can update my credentials without contacting support | <ul style="list-style-type: none"> <li>• Current password must be provided for verification</li> <li>• New password must satisfy Identity policy</li> <li>• Uses <code>userManager.ChangePasswordAsync()</code> with built-in validation</li> <li>• On success, user is notified and remains logged in</li> </ul>                                                                     | <b>Medium</b> |

## 6.3 Epic 2: Product Browsing & Discovery

| ID     | As a...            | I want to...                                              | So that...                                | Acceptance Criteria                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Priority    |
|--------|--------------------|-----------------------------------------------------------|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| US-007 | <b>Any visitor</b> | Browse a paginated grid of all active products            | I can discover what the store offers      | <ul style="list-style-type: none"> <li>• Products are displayed in an <code>auto-fill</code> CSS grid (min 260px per card)</li> <li>• Inactive products are excluded server-side via <code>p.IsActive</code> filter</li> <li>• Pagination shows 12 products per page with smart-ellipsis navigation</li> <li>• Each card shows: image, name, truncated description, price, stock badge</li> <li>• "In Stock" / "Sold Out" badges are colour-coded (green/red)</li> </ul> | <b>High</b> |
| US-008 | <b>Any visitor</b> | Filter products by category, search term, and price range | I can narrow down products to what I need | <ul style="list-style-type: none"> <li>• Filter form with: category dropdown, search text input, min/max price, sort</li> <li>• Filters apply client-side via debounced AJAX (400ms for search, 300ms for others)</li> <li>• AJAX response replaces the product grid partial without full page reload</li> <li>• URL is updated via <code>history.replaceState</code> for back-</li> </ul>                                                                               | <b>High</b> |

| ID     | As a...            | I want to...                                                               | So that...                                             | Acceptance Criteria                                                                                                                                                                                                                                                                                                                                                                                                      | Priority |
|--------|--------------------|----------------------------------------------------------------------------|--------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
|        |                    |                                                                            |                                                        | button support<br>• A loading spinner overlay is shown during AJAX requests                                                                                                                                                                                                                                                                                                                                              |          |
| US-009 | <b>Any visitor</b> | Sort products by newest, price (low-high / high-low), and name (A-Z / Z-A) | I can find products in my preferred order              | • Sort dropdown triggers AJAX reload (no page refresh)<br>• <code>price_asc</code> / <code>price_desc</code> sorts by <code>Product.Price</code><br>• <code>name_asc</code> / <code>name_desc</code> sorts by <code>Product.Name</code><br>• <code>newest</code> (default) sorts by <code>Product.CreatedDate</code> descending<br>• Server-side LINQ translates to SQL <code>ORDER BY</code>                            | Medium   |
| US-010 | <b>Any visitor</b> | View a product's full details, variants, reviews, and related products     | I can make an informed purchase decision               | • Detail page shows: large image, price, description, stock, category<br>• Variants (size/color) are displayed as selectable cards<br>• Selecting a variant updates the displayed price (base + additional)<br>• Rating summary shows 1-5 star breakdown with visual bars<br>• 4 related products from the same category are shown at the bottom<br>• "Verified Purchase" badge appears on reviews from confirmed buyers | High     |
| US-011 | <b>Any visitor</b> | Browse products by category from the homepage                              | I can discover products in a specific category quickly | • Category section on Home shows up to 6 categories in an auto-fill grid<br>• Each category card has an icon, name, and links to filtered product list<br>• Categories are cached in <code>IMemoryCache</code> for 10 minutes<br>• <code>/Products/ByCategory/{id}</code> route filters and paginates correctly                                                                                                          | Medium   |
| US-012 | <b>Any visitor</b> | View featured products on the homepage                                     | I can see the store's highlighted selections           | • Featured products ( <code>IsFeatured = true</code> ) are shown in a grid on Home<br>• Limited to 8 products, sorted by Id<br>• Each featured card includes a gold "Featured" badge<br>• Data is cached with 10-minute absolute + 2-minute sliding expiration                                                                                                                                                           | Medium   |

## 6.4 Epic 3: Shopping Cart & Wishlist

| ID     | As a...                   | I want to...                             | So that...                   | Acceptance Criteria                                                                                                      | Priority |
|--------|---------------------------|------------------------------------------|------------------------------|--------------------------------------------------------------------------------------------------------------------------|----------|
| US-013 | <b>Authenticated user</b> | Add a product to my cart with a specific | I can prepare to purchase it | • Clicking "Add to Cart" creates/updates a <code>ShoppingCart</code> record<br>• If a variant is selected, the variant's | High     |

| ID     | As a...            | I want to...                                          | So that...                                 | Acceptance Criteria                                                                                                                                                                                                                                                                                                                                                                                                  | Priority |
|--------|--------------------|-------------------------------------------------------|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
|        |                    | quantity and variant                                  |                                            | <ul style="list-style-type: none"> <li>stock is validated</li> <li>If the product already exists in the cart (same product + variant), quantity is incremented</li> <li>Cumulative quantity must not exceed available stock</li> <li>Success confirmation is shown via TempData</li> </ul>                                                                                                                           |          |
| US-014 | Authenticated user | View my cart with item details, quantities, and total | I can review my selections before checkout | <ul style="list-style-type: none"> <li>Cart page shows: thumbnail, name, variant info, quantity stepper, line total, remove button</li> <li>Inactive products are filtered out from the view</li> <li>Summary sidebar shows: subtotal, 14% tax, shipping, grand total</li> <li>Sticky sidebar follows the user on scroll (desktop)</li> <li>Empty cart shows a friendly empty-state with a "Shop Now" CTA</li> </ul> | High     |
| US-015 | Authenticated user | Increase or decrease item quantities in the cart      | I can adjust my order before checkout      | <ul style="list-style-type: none"> <li>Plus/minus buttons submit POST to <code>UpdateQuantity</code></li> <li>Quantity is bounded: min 1 (minus button disables at 1), max = stock</li> <li>Setting quantity to 0 removes the item entirely</li> <li>Stock is re-verified server-side on each update</li> </ul>                                                                                                      | High     |
| US-016 | Authenticated user | Remove individual items or clear the entire cart      | I can start over or remove unwanted items  | <ul style="list-style-type: none"> <li>Remove button (trash icon) with JavaScript confirmation dialog</li> <li>"Clear Cart" button removes all items for the current user</li> <li>Both operations call POST endpoints with <code>ValidateAntiForgeryToken</code></li> <li>Cart badge count in the header updates after any change</li> </ul>                                                                        | High     |
| US-017 | Authenticated user | Toggle products in my wishlist                        | I can save items for future consideration  | <ul style="list-style-type: none"> <li>Heart icon button on product cards toggles wishlist state</li> <li>Optimistic UI update: icon changes immediately, reverts on failure</li> <li>CSRF token is sent via <code>RequestVerificationToken</code> header</li> <li>Duplicate (user, product) is prevented by unique index in DB</li> <li>"Please sign in" toast appears for unauthenticated clicks</li> </ul>        | Medium   |

| ID     | As a...            | I want to...     | So that...                                | Acceptance Criteria                                                                                                                                                                                                                                                                                | Priority |
|--------|--------------------|------------------|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| US-018 | Authenticated user | View my wishlist | I can see all saved products in one place | <ul style="list-style-type: none"> <li>Wishlist page lists all saved products with thumbnail and price</li> <li>Quick "Add to Cart" button for each wishlist item</li> <li>Empty wishlist shows an empty-state message</li> <li>Wishlist count badge in the header updates on page load</li> </ul> | Low      |

## 6.5 Epic 4: Checkout & Payments

| ID     | As a...            | I want to...                                             | So that...                         | Acceptance Criteria                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Priority |
|--------|--------------------|----------------------------------------------------------|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| US-019 | Authenticated user | Proceed to checkout with pre-filled shipping information | I can complete my purchase quickly | <ul style="list-style-type: none"> <li>Checkout page pre-fills name, email, phone, address, city, country from user profile</li> <li>Cart items are re-loaded and totals (subtotal, 14% tax, total) are calculated</li> <li>Empty cart redirects back to cart with an error message</li> <li>Form includes all required fields: FullName, Email, Phone, Address, City, Country, PaymentMethod</li> </ul>                                                                                                | High     |
| US-020 | Authenticated user | Apply a promo code to my order                           | I can receive a discount           | <ul style="list-style-type: none"> <li>Promo code input with "Apply" button dispatches an AJAX POST to <code>ValidatePromoCode</code></li> <li>Server validates: code exists, is active, within date range, usage limit not reached, minimum purchase met</li> <li>For percentage discounts, the cap (<code>MaximumDiscount</code>) is enforced</li> <li>Discount amount and new total are displayed immediately without page reload</li> <li>Invalid/expired codes show a red error message</li> </ul> | Medium   |
| US-021 | Authenticated user | Place an order with Cash on Delivery                     | I can pay when the package arrives | <ul style="list-style-type: none"> <li><code>PlaceOrder</code> creates: <code>Order</code> (status: Pending), <code>Payment</code> (status: Pending, method: COD)</li> <li>Product stock is decremented atomically within a database transaction</li> <li>Cart items are deleted after successful order creation</li> <li>Order confirmation email is sent to the user</li> <li>User is redirected to <code>OrderConfirmation</code> page with order details</li> </ul>                                 | High     |

| ID     | As a...            | I want to...                                                                | So that...                        | Acceptance Criteria                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Priority |
|--------|--------------------|-----------------------------------------------------------------------------|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| US-022 | Authenticated user | Pay via credit card through Stripe                                          | I can pay online securely         | <ul style="list-style-type: none"> <li>• Selecting "Credit Card (Stripe)" triggers redirect to Stripe Checkout</li> <li>• Stripe session includes: order ID, total (in cents), product names</li> <li>• On success, Stripe redirects to <code>/Checkout/PaymentSuccess</code> which updates status to Paid</li> <li>• On cancellation, Stripe redirects to <code>/Checkout/PaymentCancelled</code> which sets status to Cancelled</li> <li>• If Stripe keys are not configured, a mock success path is used for development</li> </ul>                                      | High     |
| US-023 | Authenticated user | View an order confirmation page after placing an order                      | I have a record of my purchase    | <ul style="list-style-type: none"> <li>• Confirmation page shows: order ID, total, items list, payment status</li> <li>• Items are loaded via <code>OrderItems</code> with eager-loaded <code>Product</code></li> <li>• Only the owning user can view the order (authorization check via <code>UserId</code>)</li> <li>• 404 returned if the order does not belong to the current user</li> </ul>                                                                                                                                                                           | High     |
| US-024 | System             | Prevent overselling when two users checkout the same product simultaneously | I can maintain inventory accuracy | <ul style="list-style-type: none"> <li>• <code>Product</code> entity has <code>[Timestamp] byte[] RowVersion</code> concurrency token</li> <li>• <code>PlaceOrder</code> runs in a <code>for</code> loop with up to 3 retries</li> <li>• On <code>DbUpdateConcurrencyException</code>, the transaction rolls back, waits (100ms × attempt), and retries</li> <li>• After 3 failures, user sees: "Some items were just purchased by another customer"</li> <li>• Promo code <code>UsageCount</code> also uses <code>[Timestamp]</code> to prevent double-spending</li> </ul> | High     |

## 6.6 Epic 5: AI Product Assistant (Gemini Integration)

| ID     | As a...            | I want to...                                      | So that...                                           | Acceptance Criteria                                                                                                                                                                                                                                                                                                                                                                            | Priority |
|--------|--------------------|---------------------------------------------------|------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| US-025 | Authenticated user | Ask a question about a product to an AI assistant | I can get instant answers without contacting support | <ul style="list-style-type: none"> <li>• Each product card has an AI button (sparkle icon) visible on hover</li> <li>• Clicking opens a global modal with the product name + description pre-loaded</li> <li>• User types a question and presses Enter or clicks Send</li> <li>• A loading spinner with "Thinking..." is shown during API call</li> <li>• The response appears in a</li> </ul> | Medium   |

| ID     | As a...                   | I want to...                                                  | So that...                                               | Acceptance Criteria                                                                                                                                                                                                                                                                                                                                                                                   | Priority      |
|--------|---------------------------|---------------------------------------------------------------|----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
|        |                           |                                                               |                                                          | styled text box below the question input                                                                                                                                                                                                                                                                                                                                                              |               |
| US-026 | <b>Authenticated user</b> | Receive relevant answers even if my question is vague         | I can get helpful information without phrasing perfectly | <ul style="list-style-type: none"> <li>• The system prompt instructs Gemini to answer concisely and redirect off-topic questions</li> <li>• Product context (name + description) is injected into every prompt</li> <li>• Max output is 800 tokens — responses are concise</li> <li>• Temperature = 0.7 balances creativity with factual accuracy</li> </ul>                                          | <b>Medium</b> |
| US-027 | <b>Authenticated user</b> | See a friendly error message if the AI service is unavailable | I understand what happened and can try again later       | <ul style="list-style-type: none"> <li>• 429 (rate limit) returns: "AI service quota exceeded. Please try again in N seconds."</li> <li>• Network errors show: "Network error. Please try again."</li> <li>• Prompt blocked by safety filters shows: "I'm sorry, I couldn't process that request."</li> <li>• Unauthenticated users receive a 401 from the <code>[Authorize]</code> filter</li> </ul> | <b>Low</b>    |

## 6.7 Epic 6: Admin Dashboard & Management

| ID     | As a...      | I want to...                                            | So that...                                  | Acceptance Criteria                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Priority    |
|--------|--------------|---------------------------------------------------------|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| US-028 | <b>Admin</b> | View a dashboard with key business metrics              | I can monitor store performance at a glance | <ul style="list-style-type: none"> <li>• Dashboard shows: total revenue, total orders, total products, total users</li> <li>• Statistics for: pending orders, completed orders, low-stock products (&lt; 10)</li> <li>• Top 5 selling products by quantity are displayed</li> <li>• Orders grouped by payment method are shown</li> <li>• Recent 5 orders display with user name and total</li> <li>• All numbers are aggregated server-side via LINQ <code>Sum()</code> and <code>Count()</code></li> </ul> | <b>High</b> |
| US-029 | <b>Admin</b> | Manage products (create, edit, delete, toggle featured) | I can keep the product catalog up to date   | <ul style="list-style-type: none"> <li>• Product list is paginated (15 per page) with search</li> <li>• Create product form includes: name, description, price, stock, category, image upload, featured flag</li> <li>• Image upload generates a GUID filename and stores in <code>wwwroot/images/products/</code></li> </ul>                                                                                                                                                                                | <b>High</b> |

| ID     | As a...      | I want to...                              | So that...                                   | Acceptance Criteria                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Priority      |
|--------|--------------|-------------------------------------------|----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
|        |              |                                           |                                              | <ul style="list-style-type: none"> <li>Edit product pre-fills all fields and allows image replacement</li> <li>Old image is deleted from disk when a new one is uploaded</li> <li>Delete is confirmed via JavaScript prompt</li> </ul>                                                                                                                                                                                                                                                                |               |
| US-030 | <b>Admin</b> | View and manage registered users          | I can activate/deactivate or delete accounts | <ul style="list-style-type: none"> <li>User list is paginated with order count per user and assigned roles</li> <li>Role lookups are batched per role (not per user) to avoid N+1 queries</li> <li>Admin can toggle account lockout (activate/deactivate)</li> <li>Admin can delete users (with cascade cleanup of cart items)</li> <li>Self-deactivation and self-deletion are prevented</li> </ul>                                                                                                  | <b>High</b>   |
| US-031 | <b>Admin</b> | View detailed sales analytics with charts | I can track business performance over time   | <ul style="list-style-type: none"> <li>Analytics include: total orders, total revenue, average order value</li> <li>Month-over-month order and revenue growth percentages are calculated</li> <li>Top 10 selling products with quantity and revenue</li> <li>Top 10 customers by total spent</li> <li>Category performance (products sold per category)</li> <li>Daily sales chart data (last 30 days) with zero-fill for inactive days</li> <li>Monthly sales chart data (last 12 months)</li> </ul> | <b>Medium</b> |

## 6.8 Epic 7: Core Infrastructure & User Experience

| ID     | As a...            | I want to...                         | So that...                               | Acceptance Criteria                                                                                                                                                                                                                                                                                                                                                                                                                                   | Priority      |
|--------|--------------------|--------------------------------------|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| US-032 | <b>Any visitor</b> | Toggle between dark and light theme  | I can browse in my preferred visual mode | <ul style="list-style-type: none"> <li>Theme toggle button in the header switches <code>data-bs-theme</code> attribute</li> <li>Theme preference is persisted via a cookie with 1-year expiry</li> <li>FOUC is prevented: server reads the cookie and sets the attribute before HTML is sent</li> <li>All design tokens (backgrounds, text, borders, shadows) switch correctly</li> <li>Sun/moon icon toggles to reflect the current theme</li> </ul> | <b>Medium</b> |
| US-033 | <b>Any visitor</b> | Navigate the site on my mobile phone | I can shop from any device               | <ul style="list-style-type: none"> <li>Header collapses to hamburger menu at 768px</li> <li>Mobile nav includes: Home, Products, About, Contact, FAQ, and all categories</li> <li>Product grid collapses to 2 columns at 520px, 1 column at 360px</li> <li>Cart layout becomes single column below 1000px</li> </ul>                                                                                                                                  | <b>High</b>   |

| ID     | As a...     | I want to...                                                           | So that...                                     | Acceptance Criteria                                                                                                                                                                                                                                                                                                                                                                   | Priority |
|--------|-------------|------------------------------------------------------------------------|------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
|        |             |                                                                        |                                                | <ul style="list-style-type: none"> <li>Hero section hides the decorative visual ring on mobile</li> <li>Quantity stepper and buttons remain touch-friendly</li> </ul>                                                                                                                                                                                                                 |          |
| US-034 | Any visitor | Receive instant feedback when I perform actions                        | I know whether my action succeeded or failed   | <ul style="list-style-type: none"> <li>Toast notifications appear for: add/remove wishlist, add to cart, errors</li> <li>Toasts slide in from the right and auto-dismiss after 4.2 seconds</li> <li>Toasts are colour-coded (green for success, red for error)</li> <li>Each toast has a dismiss button</li> <li>Multiple toasts stack vertically</li> </ul>                          | Medium   |
| US-035 | Any visitor | View a sitemap.xml for search engine indexing                          | The store can be discovered via search engines | <ul style="list-style-type: none"> <li><code>/Home/Sitemap</code> returns valid XML with all product and category URLs</li> <li>Each URL has <code>&lt;changefreq&gt;weekly&lt;/changefreq&gt;</code> and <code>&lt;priority&gt;0.8&lt;/priority&gt;</code></li> <li>Response is cached for 86400 seconds (24 hours)</li> <li>Content-Type is <code>application/xml</code></li> </ul> | Low      |
| US-036 | Any visitor | Access privacy policy, terms of service, about, contact, and FAQ pages | I can learn about the store and its policies   | <ul style="list-style-type: none"> <li>Each page has a dedicated controller action and Razor view</li> <li>Privacy and Terms are response-cached for 3600 seconds</li> <li>These pages are linked from the footer in all four columns</li> <li>All links use <code>asp-controller</code> and <code>asp-action</code> tag helpers for correct URL generation</li> </ul>                | Low      |

## 6.9 Story Point Summary

| Epic                            | Stories   | High      | Medium    | Low      |
|---------------------------------|-----------|-----------|-----------|----------|
| 1: User Identity & Security     | 6         | 3         | 3         | 0        |
| 2: Product Browsing & Discovery | 6         | 3         | 3         | 0        |
| 3: Shopping Cart & Wishlist     | 6         | 4         | 1         | 1        |
| 4: Checkout & Payments          | 6         | 5         | 1         | 0        |
| 5: AI Assistant                 | 3         | 0         | 2         | 1        |
| 6: Admin Dashboard              | 4         | 3         | 1         | 0        |
| 7: Core Infrastructure & UX     | 5         | 2         | 2         | 1        |
| <b>Total</b>                    | <b>36</b> | <b>20</b> | <b>13</b> | <b>3</b> |

All **36 user stories** map directly to implemented code across the three-tier architecture — from Entity Framework Core entities and LINQ queries, through service-layer business logic and controller actions, to Razor views, CSS custom properties, and client-side JavaScript. The **20 high-priority stories** represent the core e-commerce workflow (auth →

browse → cart → checkout) that was implemented first, with medium and low priorities rounding out the AI assistant, analytics, and SEO features.